

2024 AMC 10A Problem 19 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10A Problem 19. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10A Problem 19

Problem:

The first three terms of a geometric sequence are the integers a , 720, and b , where $a < 720 < b$. What is the sum of the digits of the least possible value of b ?

Answer Choices:

- A. 9
- B. 12
- C. 16
- D. 18
- E. 21

Solution:

The prime factorization of 720 is $2^4 \cdot 3^2 \cdot 5$. Let $r = \frac{m}{n}$ be the common ratio of the geometric sequence, where m and n are relatively prime positive integers. If n had any prime factor greater than 5, then $b = 720r$ would not be an integer. Analogously, if m had any prime factor greater than 5, then $a = \frac{720}{r}$ would not be an integer. It follows that $r = 2^i \cdot 3^j \cdot 5^k$, where i, j , and k are (not necessarily positive) integers. Furthermore, $|i| \leq 4$, $|j| \leq 2$, and $|k| \leq 1$.

To minimize the value of b , it suffices to minimize the value of $r > 1$. Taking $r = \frac{16}{15} = 2^4 \cdot 3^{-1} \cdot 5^{-1}$ yields the sequence 675, 720, 768. To check that no lesser values of r exist, first observe that $\frac{17}{16}$ is not a possible value for r , so both n and m are greater than 17. This means that m and n must

borrow at least two prime factors each from 720, but 720 has only three distinct prime factors, so this is impossible. It follows that the least possible value of b is 768, and the requested sum of digits is $7 + 6 + 8 = (E)\boxed{21}$.

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